

An Internship Report
Of
CYBERSECURITY & WEB APPLICATION SECURITY
ASSESSMENT INTERNSHIP

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech (CSE)



By

Roll Number of Student: 25SCS1003002279

Name of Student: Singamsetty Shanmukha

Batch: 2025-2029

CANDIDATE’S DECLARATION

I, Singamsetty Shanmukha do hereby declare that the internship report titled **“CYBERSECURITY & WEB APPLICATION SECURITY ASSESSMENT INTERNSHIP”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Student Name: **Singamsetty Shanmukha**

Roll No: **25SCS1003002279**

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the cybersecurity internship program team, **InternNova**, for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of **Cybersecurity & Web Application Security**. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:

Student Name: **Singamsetty Shanmukha**

Roll No: **25SCS1003002279**

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3. INTERNSHIP COMPLETION CERTIFICATE



Certificate No.: IN/26/CS9573

CERTIFICATE

OF INTERNSHIP

This is proudly presented to

Singamsetty Shanmukha

In recognition of the successful completion of the 6-Week Cyber Security Internship Program at InternNova, conducted from 01 Aug 2026 to 11 Sep 2026. Exceptional dedication and professionalism were demonstrated in completing all assigned tasks and projects, reflecting strong technical aptitude and commitment to professional growth.



AVYA NAND
Founder & CEO



Registered on AICTE Internship Portal
MSME Registered Entity
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Issue Date: 12 Sep 2026

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3.1 INTERNSHIP OFFER LETTER



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INTERNSHIP OFFER LETTER

Date: 19/07/2026

Offer Letter ID: INOL/26/CS/024

Dear Singamsetty Shanmukha,

We are delighted to offer you the position of Cyber Security Intern at InternNova. This Virtual Internship Program is designed to provide practical industry exposure, project-based learning and hands-on experience that will strengthen your technical and professional skills.

Internship Details	Information
Domain	Cyber Security
Mode	Virtual
Duration	6 Week
Start Date	01 August 2026
End Date	11 September 2026
Status	Selected

IMPORTANT INFORMATION

Internship tasks, project assignments, onboarding instructions and communication channels will be shared separately through official InternNova communication channels before commencement.

CERTIFICATE ELIGIBILITY

To receive the Internship Completion Certificate, the intern must complete assigned tasks/projects, follow internship guidelines, maintain professional conduct and submit required work within deadlines.

We congratulate you on your selection and wish you a successful learning journey with InternNova.

Best Regards,

Aniya Nand
Founder & CEO
InternNova



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Verify: www.internnova.co.in/#verify

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the 6-week internship undertaken in the domain of **Cybersecurity and Web Application Security**. The internship was carried out as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to bridge the gap between theoretical computer networking, security concepts and practical security awareness.

Over the course of the programme, exposure was gained to foundational networking architectures, threat intelligence concepts, vulnerability identification workflows, and security assessment methodologies. The internship culminated in conducting structured security assessments on a designated test application (*testasp.vulnweb.com*), analyzing potential security gaps, and compiling a final security evaluation report.

4.2 Organization Profile

The internship was conducted through a specialized Cybersecurity Track structured to deliver guided learning across foundational offensive and defensive security operations. The track provides a structured platform combining theoretical modules, conceptual assessments, vulnerability testing frameworks, and web application security analysis. The program emphasizes industry-aligned cybersecurity fundamentals, threat awareness, and risk analysis methodologies designed to enhance engineering students' technical domain understanding.

4.3 Problem Statement

Modern web applications and network infrastructures face continuous security risks due to misconfigurations, weak input validation, and unpatched system vulnerabilities. While organizations deploy web interfaces for public and internal workflows, many systems remain vulnerable to threats such as SQL Injection (SQLi), Cross-Site Scripting (XSS), phishing attacks, and credential weakness.

Understanding security concepts in theory is insufficient; evaluating real-world web environments to identify potential attack vectors is essential. The problem addressed during this internship was to analyze target web application structures, map attack surfaces, recognize common vulnerabilities, and document recommended security practices to help safeguard web-based systems.

4.4 Project Objectives

- To understand foundational computer networking principles, including IP & MAC addressing, OSI layer functions, and protocol operations.
- To learn threat intelligence principles, social engineering mechanisms, and phishing attack analysis.
- To study network scanning principles, service discovery concepts, and traffic monitoring basics.
- To perform vulnerability assessments on authorized test web applications to identify common web security risks such as SQL Injection and XSS.
- To understand risk scoring methodologies and document security findings into structured assessment reports.

4.5 Scope of the Project

The scope of this internship encompasses the following learning domains:

- **Networking & Protocols:** Understanding IP/MAC structures, routing fundamentals, packet flow, and browser-server interaction.
- **Threat & Phishing Awareness:** Analyzing email header structures, social engineering techniques, and common malware delivery mechanisms.
- **Web Application Security Assessment:** Analyzing designated test targets (e.g., *testasp.vulnweb.com*) for input validation issues, parameter handling, and basic web security flaws.

4.6 Technologies and Tools Used

- **Operating Systems:** Linux (Debian/Fedora/Arch environment) & Windows

- **Network & Scanning Tools:** Nmap, Wireshark, Netcat
- **Web & Browser Tools:** Browser Developer Tools (Inspect Element, Network Tab, Storage Inspector)
- **Languages & Scripting:** Python, Shell Scripting (Bash)
- **Diagrams & Documentation Tools:** Excalidraw, Canva, Markdown
- **Target Assessment Environments:** Authorized test web platforms (e.g., *testasp.vulnweb.com*)

4.7 System Architecture

The technical architecture and assessment workflow followed during the web application security evaluation can be broken down into the following key components:

- **Target Application Layer:** Represents the external target web platform (*testasp.vulnweb.com*) hosted on a web server, serving HTTP/HTTPS web forms, input parameters, and dynamic pages.
- **Client / Workstation Layer:** The local security workstation running a Linux environment, used to initiate requests and observe server responses.
- **Inspection & Analysis Layer:** Utilizing standard Browser Developer Tools (Inspect Element, Network Tab, and Storage Inspector) to monitor request/response headers, analyze HTML input fields, inspect cookie properties, and track web traffic flows.
- **Manual Input Testing:** Manually testing application form fields, query parameters, and URL inputs to evaluate input handling and identify basic security risks such as SQL Injection and Cross-Site Scripting (XSS) concepts.
- **Documentation & Reporting Layer:** Structuring observed security behaviors, potential attack vectors, and risk evaluations using visual design tools (Excalidraw/Canva) and Markdown for final report compilation.

4.8 Methodology

The internship followed a structured, week-by-week methodology combining conceptual learning, practical modules, weekly coursework, and a final assessment submission:

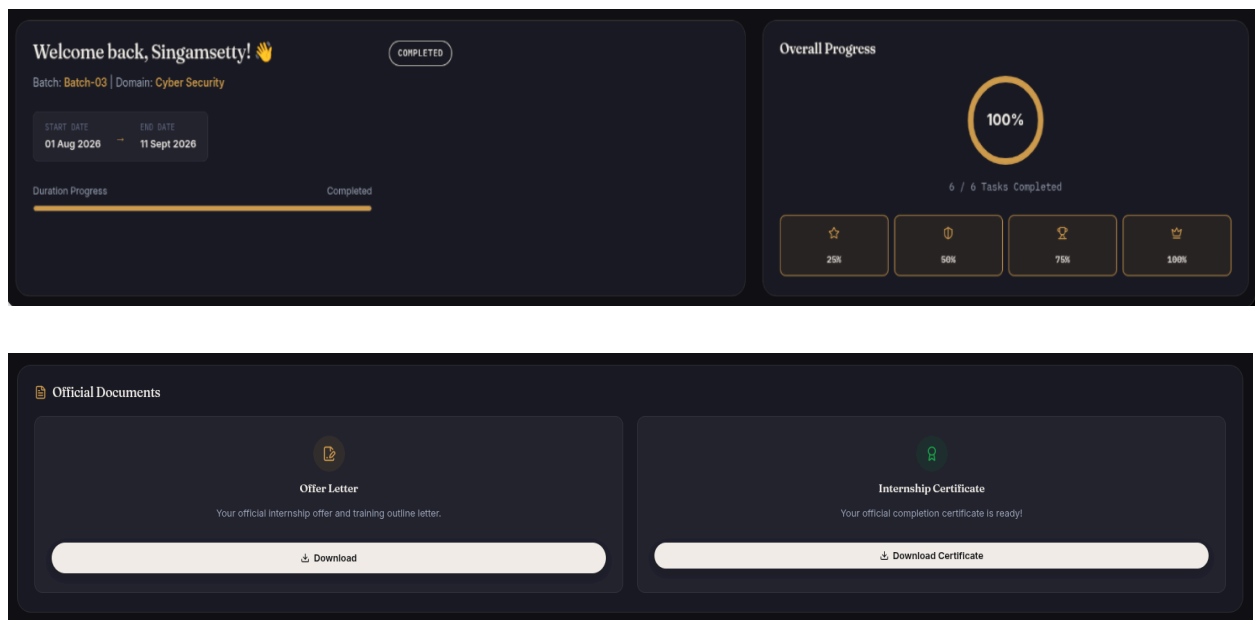
Week	Module	Description
1	Computer Networking & Protocol Analysis	Analyzing OSI/TCP-IP reference models, IP subnetting, MAC mapping mechanics, packet encapsulation, and inspecting stateless vs. stateful protocol behaviors (HTTP/HTTPS, DNS, ARP).
2	Threat Intelligence & Social Engineering Vectors	Examining threat actor tactics, phishing mechanics, SMTP/email header analysis, payload delivery vectors, and defensive security awareness.
3	Network Reconnaissance & Port Scanning	Studying active/passive footprinting principles, network discovery workflows, host discovery, TCP SYN/Connect scanning concepts, and Nmap service version fingerprinting.
4	Web Application Vulnerability Assessment	Auditing target web application structures (<i>testasp.vulnweb.com</i>), analyzing parameter handling, evaluating client-side validation bypasses, and studying OWASP Top 10 concepts (SQL Injection, Stored/Reflected XSS).
5	Vulnerability Analysis & Risk Quantification	Evaluating identified attack surfaces, mapping security misconfigurations, classifying threat severity using risk assessment metrics, and detailing remediation patches.
6	Security Evaluation & Final Capstone Report	Built a Python-based Security Log Analyzer & Suspicious Activity Detector, compiling vulnerability findings, and formulating the final technical report.

4.9 Expected Outcomes

- A clear conceptual understanding of computer networking principles and internet security protocols.
- Ability to recognize common web application vulnerabilities such as SQL Injection and Cross-Site Scripting.
- Practical familiarity with browser-based developer tools for analyzing web application responses and form fields.
- Preparation of structured security assessment reports detailing observed vulnerabilities and general mitigation recommendations.

4.10 Certificates of Completion and Communication Proof

The official Internship Completion Certificate, along with assignment communication proofs, are attached in Chapter 3 and below:



5. BIBLIOGRAPHY/REFERENCES

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